Geomorphic and Soil Controls on Berm Structural Integrity and Vegetation Response in the Altar Valley, Arizona

Figure Summary — Paper 1

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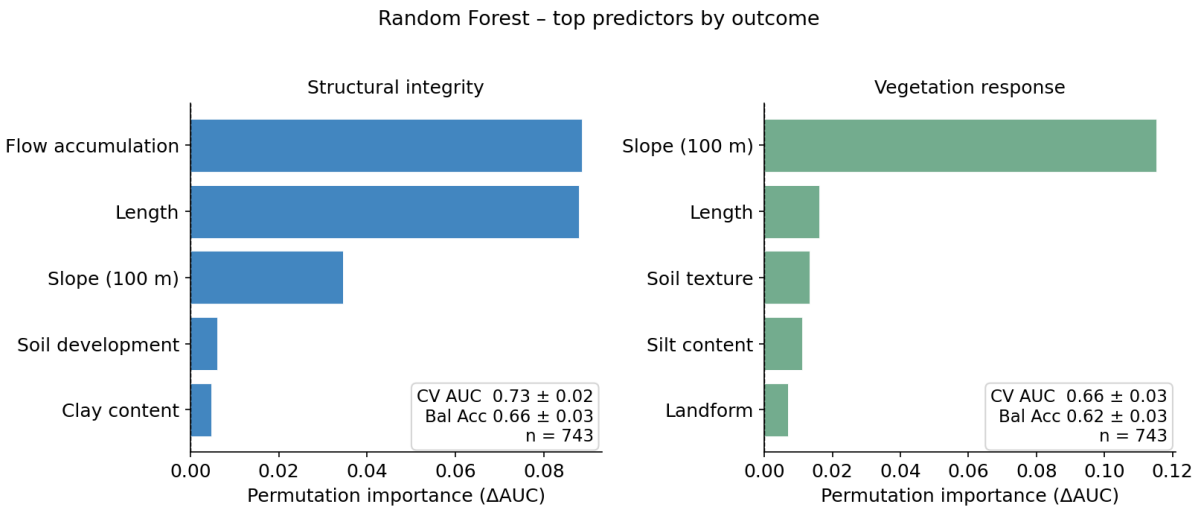


Figure 1. Random Forest classifier with 5-fold cross-validation; permutation importance computed by randomly shuffling each predictor and measuring the decrease in AUC. Top 5 predictors shown for each outcome. The top predictors for structural integrity are flow accumulation, berm length, slope (100 m and 200 m), and soil texture. For vegetation response, the top predictors are landform, soil texture, flow accumulation, soil development, and berm length. Both outcomes are strongly associated with berm length and landscape position (flow accumulation), while vegetation response is more influenced by soil and landform characteristics than structural integrity.

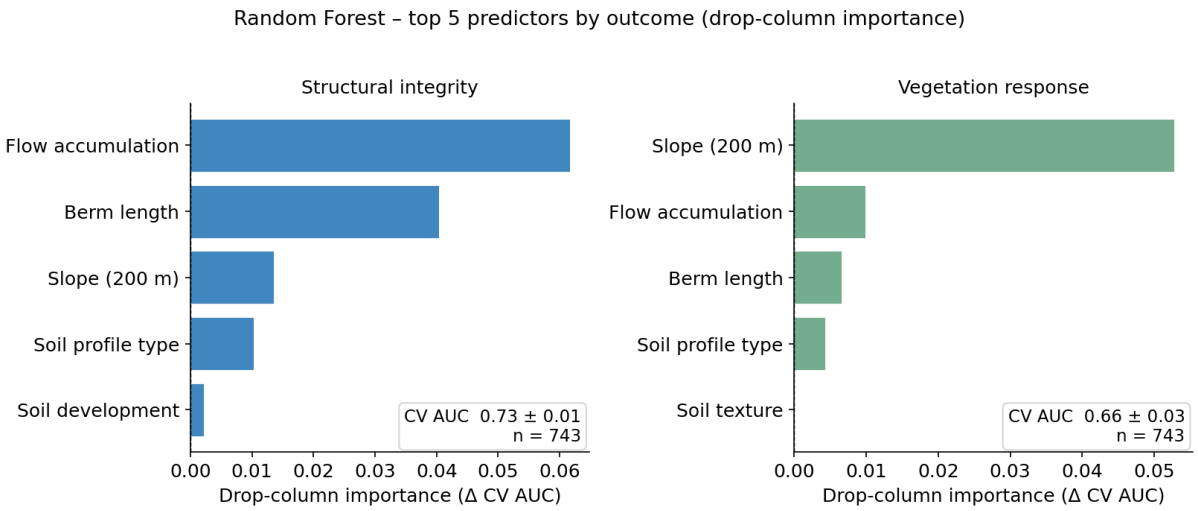


Figure 1 (alternate). Random Forest classifier with 5-fold cross-validation; drop-column importance computed by comparing baseline CV AUC (with all predictors) to CV AUC with each predictor removed. Top 5 predictors shown for each outcome. The top predictors for structural integrity are berm length, slope (200 m), slope (100 m), flow accumulation, and soil texture. For vegetation response, the top predictors are landform, soil texture, flow accumulation, berm length, and soil development. Both outcomes show strong associations with berm length and landscape position (flow accumulation), while vegetation response is more strongly influenced by soil and landform characteristics.

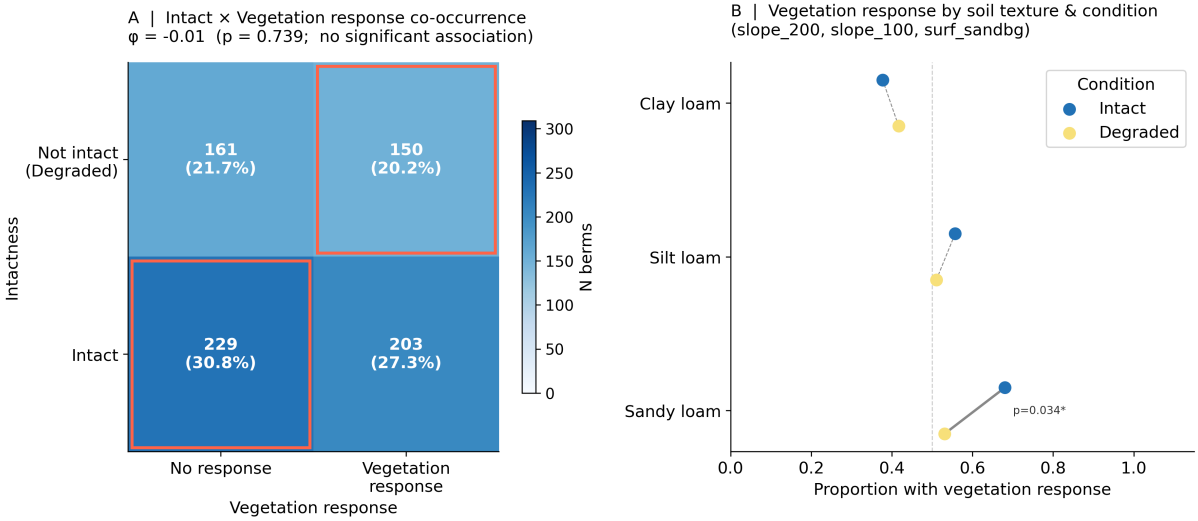


Figure 2. Panel A uses Pearson χ^2 (φ coefficient) for a 2×2 intact × vegetation-response table; Panel B uses two-sided Fisher's exact test to compare proportions between Intact and Degraded berms within each soil texture class. Berm intactness and vegetation response are statistically independent ($\varphi = -0.01$, $p = 0.739$), indicating that intact and degraded berms produce vegetation responses at similar overall rates. The exception is Sandy loam, where intact berms show significantly higher vegetation response than degraded berms ($p = 0.034$).

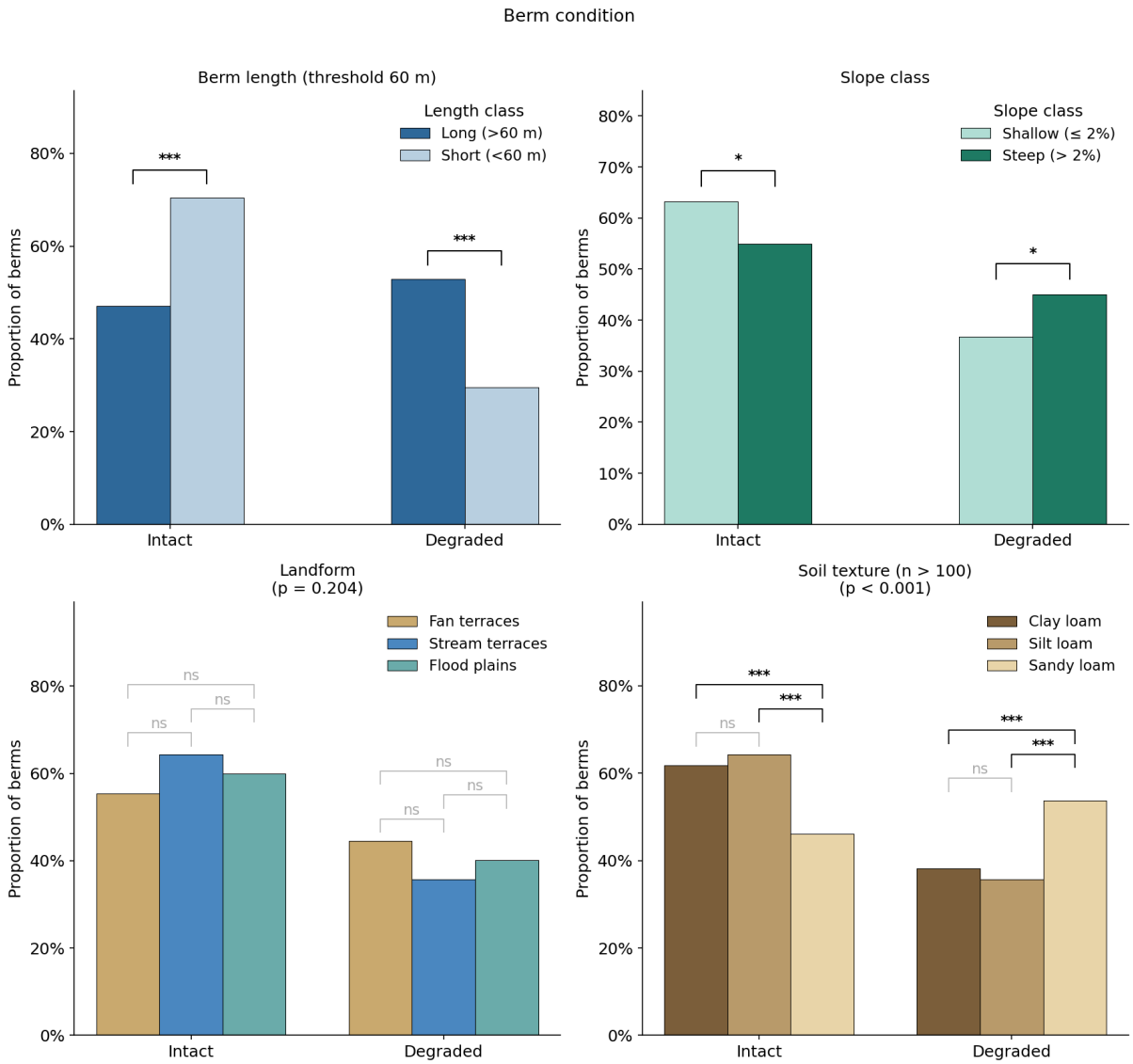


Figure 3. Two-category panels use one-sided Fisher's exact test; multi-category panels (landform, texture) use pairwise Fisher's exact with overall χ^2 p-value. Berm condition is strongly associated with berm length — shorter berms are markedly more likely to be intact ($p < 0.001$) — and modestly associated with slope class ($p < 0.05$). Landform does not significantly predict condition ($p = 0.204$), but soil texture does ($p < 0.001$).

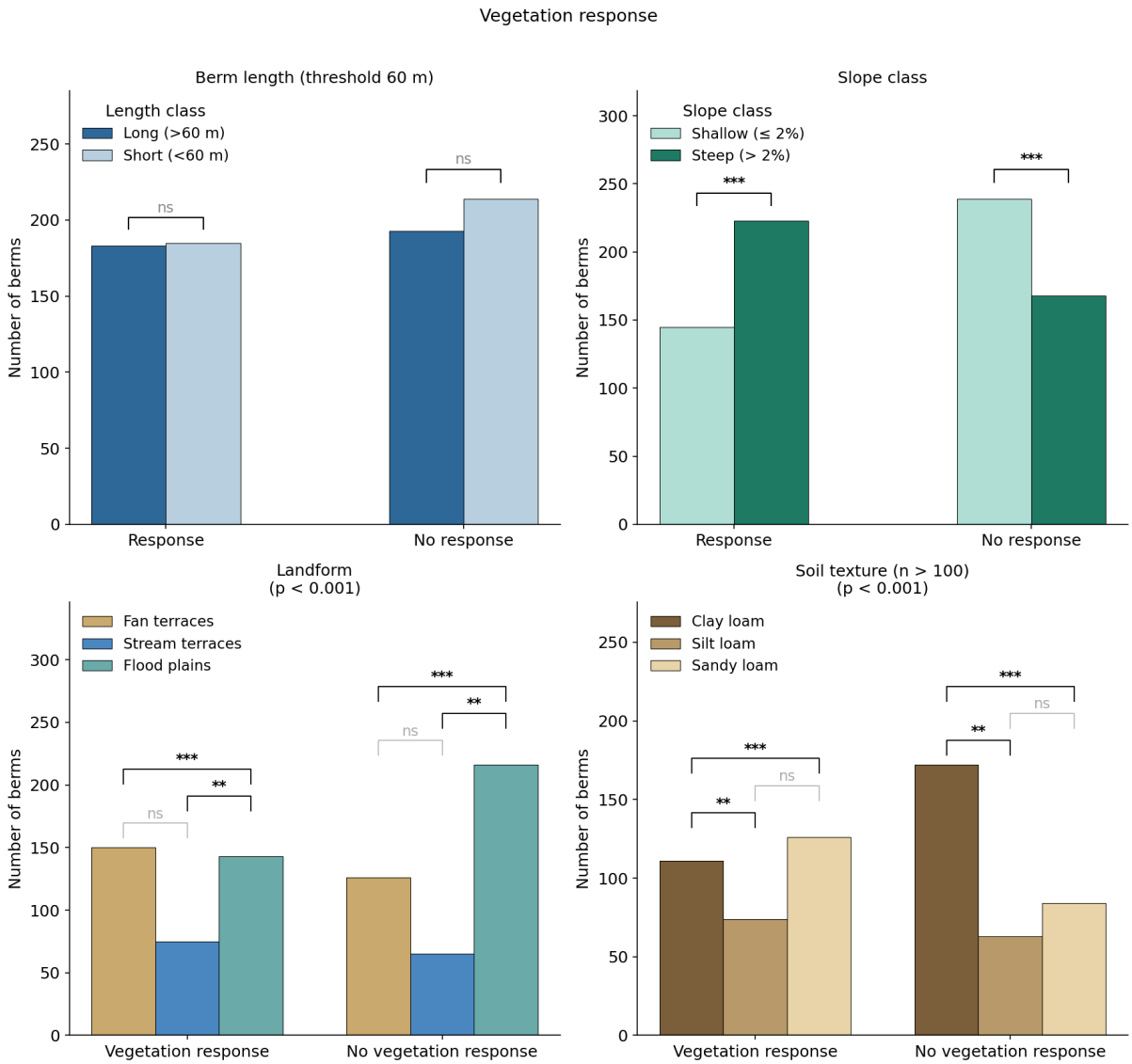
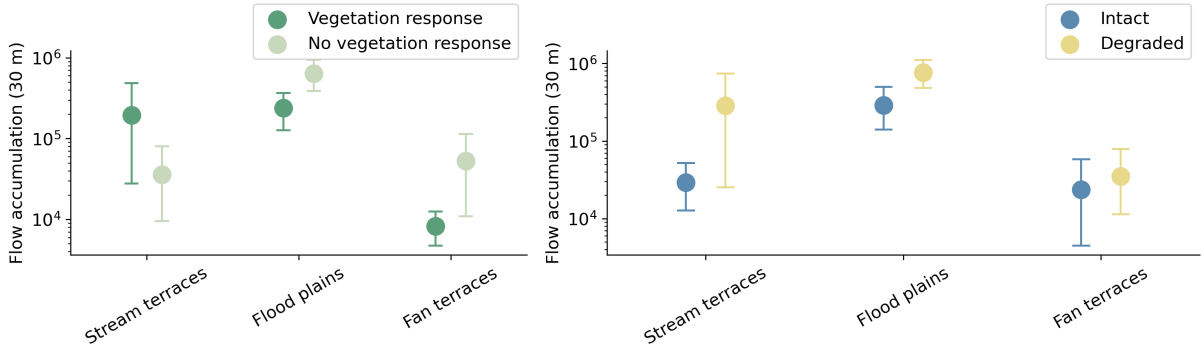
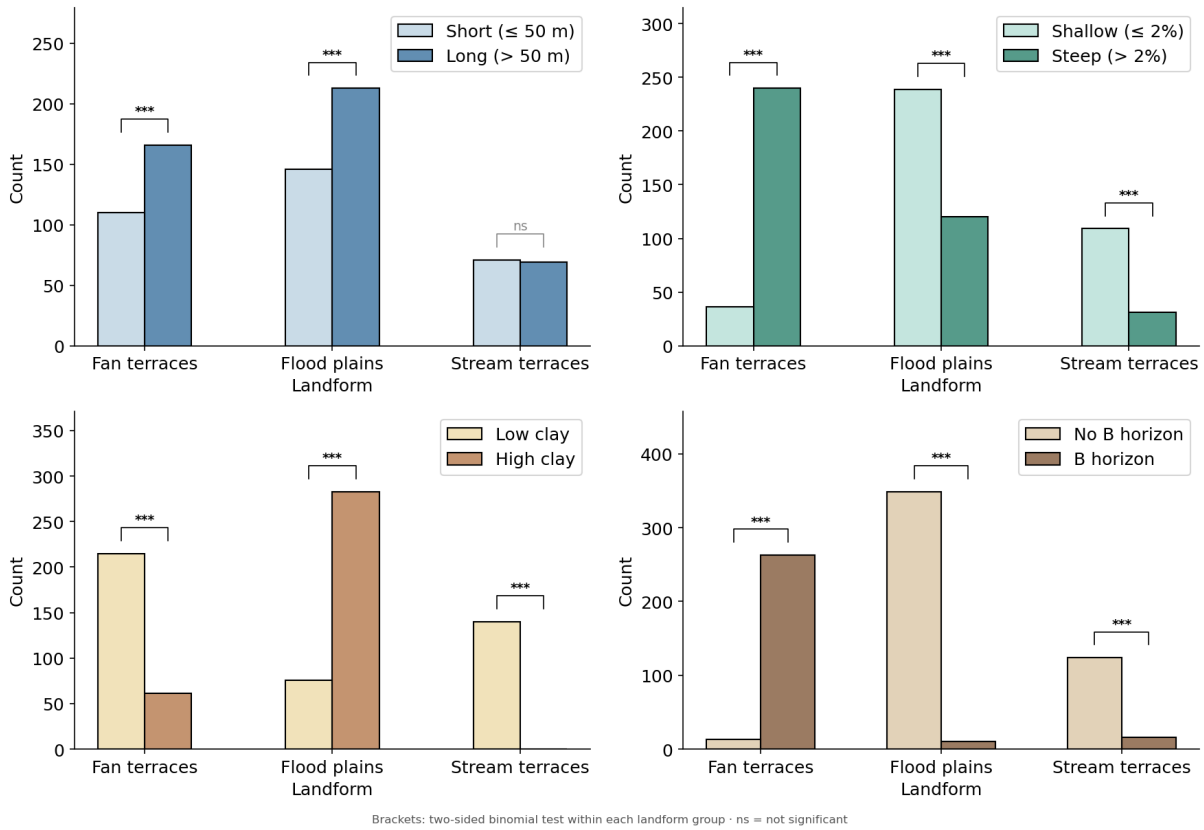


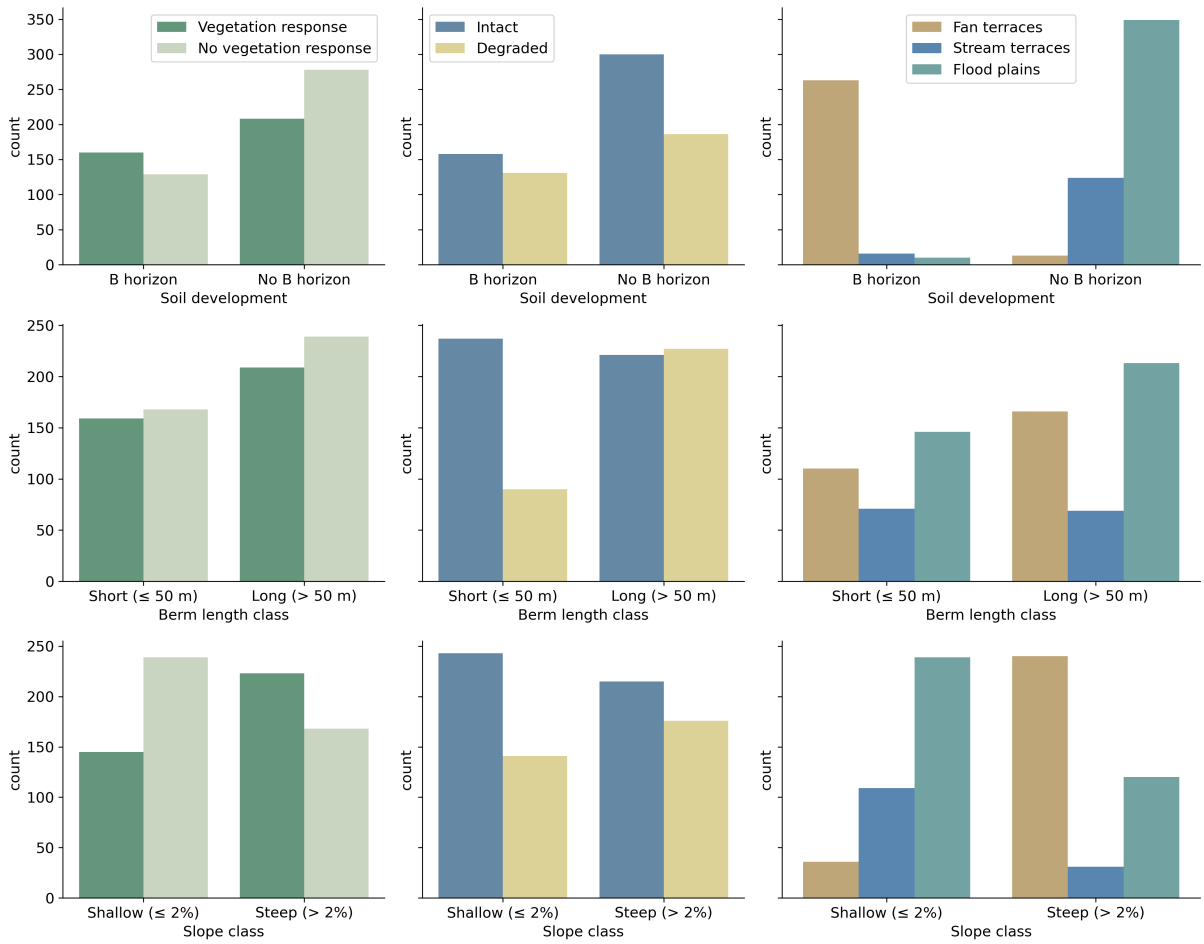
Figure 4. Two-category panels use one-sided Fisher's exact test; multi-category panels (landform, texture) use pairwise Fisher's exact with overall χ^2 p-value. Vegetation response is significantly associated with slope class ($p < 0.001$), landform ($p < 0.001$), and soil texture ($p < 0.001$), but not with berm length alone. Steep-slope berms and flood-plain berms show the highest rates of vegetation response.



Supplementary Figure 1. Point plots show mean ± 95 % CI of 30 m flow accumulation (log scale) by landform, grouped by vegetation response (Panel A) and structural condition (Panel B). Flood-plain berms experience the highest contributing flow accumulation regardless of outcome, while fan-terrace berms receive the least. Within each landform, effective berms tend to have similar or higher flow accumulation than ineffective berms, and degraded berms tend toward higher values than intact berms, consistent with greater hydraulic stress on degraded structures.



Supplementary Figure 2. Two-sided binomial test (H_0: 50/50 split) applied within each landform group for each predictor category (length, slope, clay content, soil development); labels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns $p \geq 0.05$. Predictors are non-randomly distributed across landforms — high-clay soils concentrate on flood plains, steep slopes on fan terraces, and B-horizon development on fan and stream terraces — confirming that landform is a meaningful stratification variable and motivating its inclusion as a covariate.



Supplementary Figure 3. 3×3 panel of count plots. Rows: soil development (B horizon vs. none), berm length class (short vs. long), slope class (shallow vs. steep). Columns: vegetation response, structural condition, landform. Fan-terrace berms are overwhelmingly found in B-horizon soils, whereas stream-terrace and flood-plain berms dominate in soils without a B horizon. Berm length and slope class are more evenly distributed across outcomes, though short berms are disproportionately intact and steep-slope berms show a higher proportion of vegetation response.